



New Relic for Students

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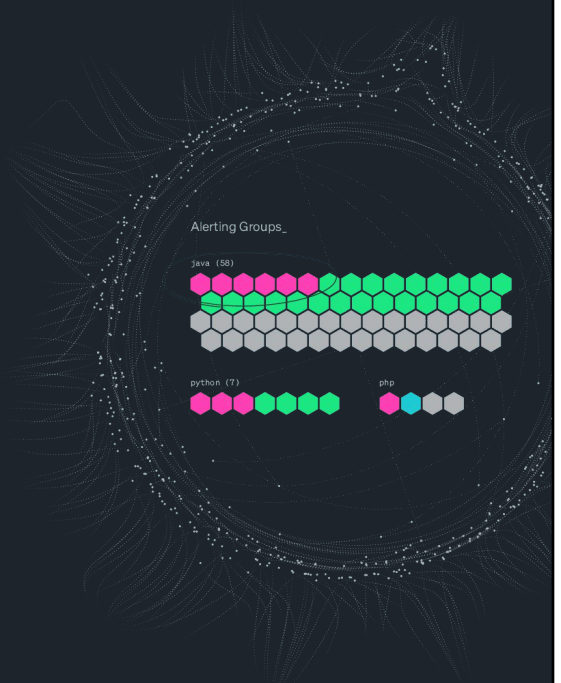
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New Relic Alerts Fundamentals

Phil Weber
Technical Instructor

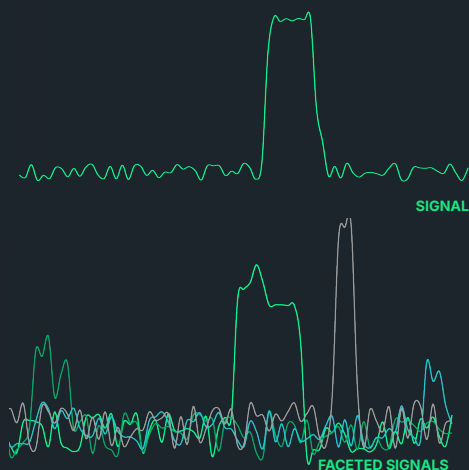
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Agenda

- 01 Alerts terminology and concepts
- 02 Lab: Creating alert conditions and policies
- 03 Lab: Configuring workflows and destinations
- 04 Managing issues
- 05 Muting rules
- 06 Additional resources

What is a "Signal"?



Acquisition → Detection → Noise reduction → Routing → Notify

Signal

- **Any data** being sent to New Relic
- Result of an **NRQL query** against incoming data
- If you can **query it you can alert** on it!

Faceted signal

- Faceting the query results in **multiple signals**
- Each signal can trigger separate alerts

Signals to alert on

Some ideas:

- Host metrics: CPU, memory, disk space, restarts
- APM golden metrics: Latency, throughput, errors
- Key transactions
- Synthetic journeys: Success rate, duration
- Log data: Error messages, business data
- Business metrics: Customers, sales



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Conditions

Signal acquisition and evaluation



Some examples

Signal	Threshold
% Error rate	% error > 5%
Average basket value	median(value) < \$20
Login attempts per user	uniqueCount(user) > 10

Acquisition Detection Noise reduction Routing Notify

Signal query

Defined by NRQL query including optional WHERE and FACET clauses.

Guided wizard available to help build.

**Additional non-NRQL conditions soon to be deprecated*

Threshold

Value, which if breached by signal for a defined time period, will trigger an incident to be opened.

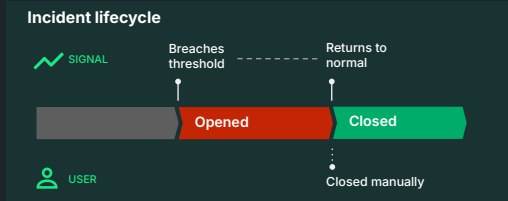
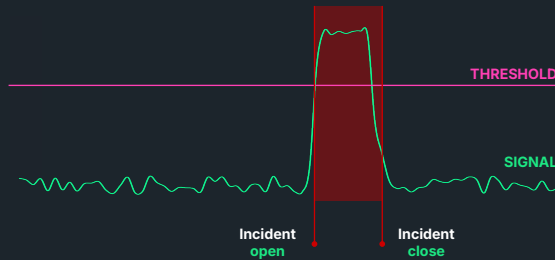
A threshold can be set for both **warning** and **critical**

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Incidents

Signal breach detection



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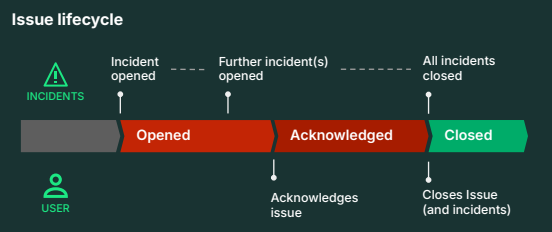
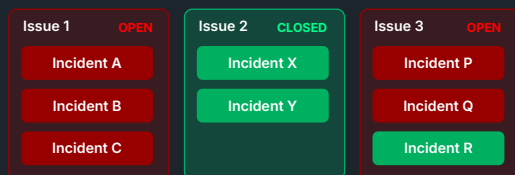
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Acquisition Detection Noise reduction Routing Notify

- Incidents **open** when a **signal breaches the threshold** defined in a condition.
- Incidents **close automatically** when the signal is no longer in breach.
- There will be some **latency** between signal breaching and incident opening based on condition settings.
- Incident will open for **each signal facet**
- Incidents can be **manually closed** by user

Issues

Incident management and noise reduction



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[Documentation about issues](#)

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Acquisition Detection Noise reduction Routing Notify

Incidents are the symptoms of a larger problem (the issue).

Issues **group incidents together**, reducing noise and driving notification workflows.

- Issues are **opened when incidents open**
- Issues can **contain multiple incidents**
based on policy preference or correlation decisions
- Issues **close automatically** when all contained incidents have closed or if inactive for defined period
- Issues can be **manually closed**, which close all contained incidents.

Policies

Predetermined incident grouping

Acquisition

Detection

Noise reduction

Routing

Notify

FEWER
ISSUES

1

Per
Policy



Only one issue will be open at a time for the **entire policy**.

- ❑ Requires immediate action and closing the issues to be effective

2

Per
Condition



One issue will be open at a time for **each condition** in your policy.

- ❑ Useful for policies containing conditions that focus on entities that perform the same job

3

Per
Incident



An issue will be created for **every incident** of **each condition** in your policy.

- ❑ Useful if you need to be notified of every violation or if you have an external system where you want to send alert notifications

MORE
ISSUES

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[Documentation on alert policies / issue preference](#)

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Lab: Alert conditions and policy

Perform the following tasks in NRU Training account:

1. Go to *Alerts > Alert Conditions* and create two alert conditions:
 - Create a critical incident if any APM service in the *Production* environment has a response time greater than 500 ms for at least 5 minutes
 - Create a critical incident if the page load time for the New Relic Pet Clinic Browser application is greater than 2 seconds for at least 5 minutes
2. Add the two conditions to a new alert policy. Give it a name you will recognize, such as *[Your Name] Policy*

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Lab: NRQL alert conditions

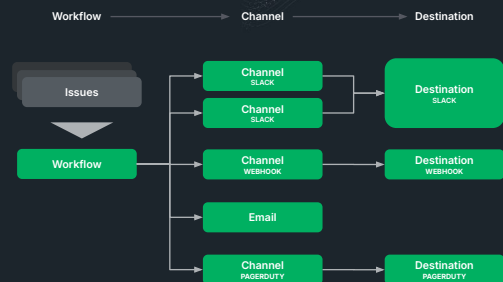
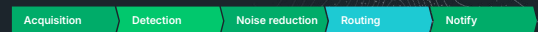
Perform the following tasks in NRU Training account:

1. Build a query that returns the average CPU utilization for any host running the *New Relic Pet Clinic* APM application
2. Create a *critical* incident if the query returns a value greater than 50% for at least 5 minutes
3. *Optionally*, add a threshold that creates a *warning* incident if CPU utilization is greater than 20% for at least 3 minutes
4. Add the new condition to the alert policy you created earlier

Workflows

Notification triage and routing

- ❑ Control **when** you want to receive notifications about issues
- ❑ **Notify correct teams** based on issue context
- ❑ Channels offer comprehensive **payload templating** options
- ❑ **Enrich notifications** with additional New Relic data



Channels allow **multiple different message payloads** to be sent to **multiple destinations**

Destinations

Notification dispatch

Destination inform **people** or **third party services** about issues state change.

Supported destinations:

- ☐ Email & mobile push
- ☐ Native: Jira*, ServiceNow*, PagerDuty*, Slack
- ☐ Webhook: e.g. OpsGenie, MS Teams, etc..
- ☐ AWS EventBridge

Destinations **setup once per account** and can be used by multiple workflows.



servicenow



PagerDuty

*two way integrations

Lab: Workflows and destinations

Perform the following tasks in **NRU Training** account:

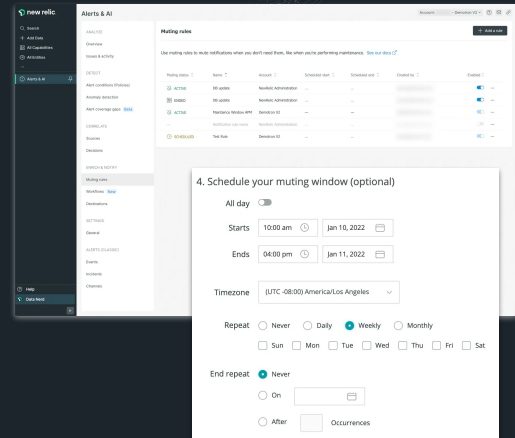
1. Select *Alerts > Workflows* and add a workflow
2. Give it a name you will recognize, such as *[Your name] workflow*
3. Under **Filter data**, create a Basic or Advanced filter that matches your high CPU condition
4. Under **Notify**, select *Email* and enter your email address
5. *Optional*: Click *Test workflow*. Did you receive an email?
6. Click the *Activate workflow* button in the lower-right corner

Muting rules

Temporarily mute notifications



- ❑ Mute **issue notifications**, for instance when **performing maintenance** or known **down periods**
- ❑ Issues are **still generated and queryable**, they just don't send notifications.
- ❑ **Filter incidents** by attribute, can target multiple workflows and destinations.
- ❑ Muting rules can be **scheduled and repeatable**



Resources

- [Alerts documentation](#)

Blog posts:

- [Best practices for fixing your alerts](#)
- [Alert! Best practices for alert quality](#)

Documentation links:

- [Alert conditions](#)
- [Issues and incident management](#)
- [Alert policies](#)
- [Issue incident grouping preference](#)
- [Workflows in New Relic](#)
- [Notification destination integrations](#)



Data Bytes

Short how-to videos covering a variety of topics

- [The new alert condition creation form](#)
- [Never miss an alert with New Relic's mobile app](#)
- [Setting muting rules for workflows](#)
- [How to set up webhook destinations for workflows](#)
- [Set up sliding window aggregation in your alert conditions](#)
- [Alert aggregation methods explained](#)

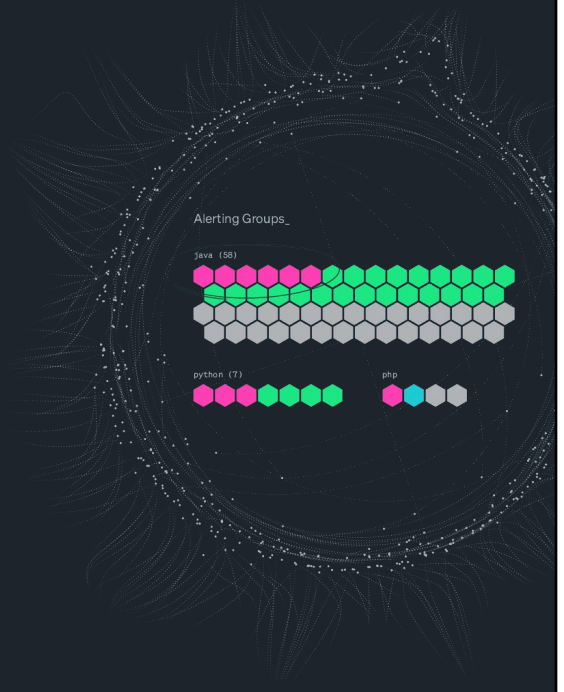
Destination setup guides

- [How to configure workflow email destinations](#)
- [Set up a JIRA workflow destination](#)
- [Send notifications to Slack using workflow destinations](#)
- [How to configure workflow Pagerduty destinations](#)
- [How to configure ServiceNow workflow destinations](#)

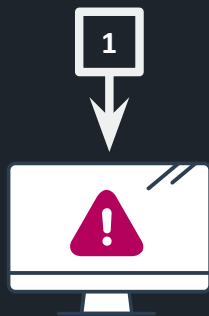
Digital Experience Monitoring

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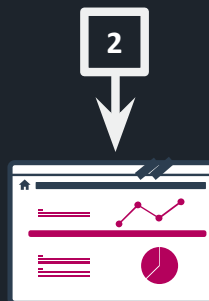
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RUM doesn't always tell the whole story



Many variables affect performance



Users must visit page in order to get data

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Synthetics

Proactive monitoring

Proactive
Find problems
before users
see them



Consistent,
predictable
Good for alerting

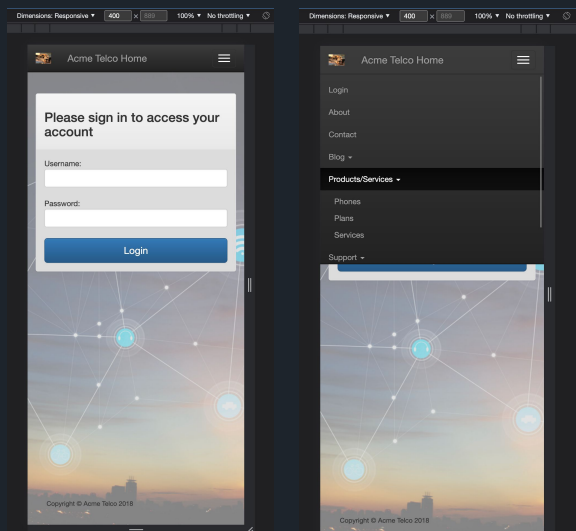
Synthetics monitor types

- Broken links monitor
- Certificate check monitor
- Ping monitor
- Simple browser
- Step monitor
- Scripted browser
- API test

Lab: Creating ping and simple browser monitors

1. Create an **Availability (Ping)** monitor to check the URL of your choice from 3 locations
2. Create a **Page load performance (Simple browser)** monitor for the same URL

Device Emulator



Use your tests to check navigation on different screen sizes and resolutions

Lab: Creating a step builder monitor

Create a **Step builder** monitor that performs the following steps:

1. **Navigate** to <http://webportal.telco.nrdemo.com/>
2. **Click Element** Login
3. **Type Text** in element input[name="username"], your user name
4. **Type Text** in element input[name="password"], any password
5. **Click Element** input.btn-primary
6. **Assert Element** //a[text()="<your user name>"] is visible

Lab: Creating a scripted API test

Create an **Endpoint availability (Scripted API)** monitor and paste the following code:

```
const assert = require('assert');

$http.get('https://csrng.net/csrng/csrng.php?min=0&max=255',
  // Callback
  function (err, response, body) {
    assert.equal(response.statusCode, 200, 'Expected a 200 OK response');
    var value = body[0].random;
    console.log('Returned value: ' + value);
    assert.equal(value < 128, true, 'Expected value less than 128');
  }
);
```

Private locations

- Allow you to monitor internal sites, add new locations
- Can install custom Node modules
- Each private location is a collection of private "minions" (Docker containers)

<https://docs.newrelic.com/docs/synthetics/synthetic-monitoring/private-locations/private-locations-overview-monitor-internal-sites-add-new-locations/>

Reporting on Synthetic tests

- Service levels
 - Deploy success rate
 - Service stability
- Dashboards
 - Success rate of individual monitors
 - Which monitors or locations are failing
 - Monitoring site performance

Lab: Synthetics dashboard

Create a dashboard with the following queries:

- `SELECT percentage(count(result), WHERE result = 'SUCCESS')
AS 'Success Rate' FROM SyntheticCheck WHERE monitorName = 'Your
Monitor Name'`
- `SELECT count(*) FROM SyntheticCheck FACET error`
- `SELECT average(duration) FROM SyntheticCheck FACET locationLabel`

Lab: Synthetics alert conditions

- Create a Synthetics alert condition that will create a critical incident if two or more locations fail at the same time
- Create the following NRQL conditions:
 - Static threshold: ≥ 1 in 15 minutes
`SELECT count(result) FROM SyntheticCheck WHERE result = 'FAILED'`
 - Anomaly threshold: current value deviates from typical performance
`SELECT max(firstContentfulPaint) FROM SyntheticRequest`